

VAYUSENSE

Autonomous Environmental Command Infrastructure

SYSTEM METADATA:

Project Title:	VayuSense (Command Tier 4 Platform)
Architecture Version:	v1.2.0 (Stable Production Build)
Target Municipalities:	277 Corporations across all 28 States + Delhi UT
Ingestion Feasibility:	Real-time Ingestion (PM2.5, PM10, SO2, NO2, O3, CO, NH3)
Release Date:	7/22/2026

VayuSense utilizes a multi-cloud production architecture incorporating Server-Sent Events (SSE) telemetry data, ESA satellite plumes, and autonomous CrewAI agentic command workflows to dynamically identify and issue statutory compliance enforcement orders across dense Indian cities.

AUTHORIZATION SYSTEM SEAL:

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Digitally verified with SHA-256 statutory log audit system.

SYSTEM ARCHITECTURE BLUEPRINT



FIGURE 2.0: VAYUSENSE INTER-LAYER COMMUNICATION TOPOLOGY

- The client triggers scenario projections via REST POST payloads to the Render endpoint.
- CrewAI agents query PostGIS database indexes to cross-reference coordinates and emit blameworthiness.
- Live telemetry streams continuously bypass the database via Server-Sent Events (SSE) loops.

COMPONENT CONFIGURATION MATRIX

SERVICE LAYER	DEPLOYMENT HOST / ENGINE	URL DIRECTORY REFERENCE
Frontend App	Vercel Serverless Hosting	vayusense-nu.vercel.app
AI Inference Server	Render Python Web Service	vayusense-backend.onrender.com
Interactive Docs	FastAPI OpenAPI Autogen	...backend.onrender.com/docs
Database Engine	Supabase / Neon Cloud Postgres	db.supabase.co:5432
Developer Portal	Next.js Static Component	...nu.vercel.app/dashboard/docs

ACTIVE EMISSION FACTOR MATRIX SPECIFICATIONS:

%² INDUSTRIAL & POWER GENERATION (50% WEIGHT)

- Target Pollutants: Sulphur Dioxide (SO2), Nitrogen Oxides (NOx), Fine Particulates (PM2.5), Lead (Pb).
- Visual Accent: Yellow solid triangles mapped over designated municipal factory grid cells.
- Compliance Trigger: Automatic show-cause PDF notice dispatch to plant inspector desk on AQI > 200.

%ĭ VEHICULAR & TRANSPORTATION EXHAUST (30% WEIGHT)

- Target Pollutants: Nitrogen Dioxide (NO2), Black Carbon carbon particulates.
- Visual Accent: Cyan circle-line corridors highlighting high-density highway and street grid patterns.
- Compliance Trigger: Direct recommendations to reroute logistics diesel fleets via automated dispatches.

% CONSTRUCTION & DEPOSITION DUST (20% WEIGHT)

- Target Pollutants: Particulate Matter 10 (PM10 Silt Dust).
- Visual Accent: Purple square boundaries isolated around active demolition and development zones.
- Compliance Trigger: Mandatory deployment of water-fogging tankers and dry silt sweeping trucks.